

Breadth-First Search (BFS) Using Python

Aim

To write a Python program to implement the **Breadth-First Search (BFS)** algorithm for traversing a graph level by level starting from a given source node.

Algorithm

1. Start the program.
2. Define the graph using an adjacency list.
3. Create an empty set to store the visited nodes.
4. Create a queue and insert the starting node.
5. Remove the front node from the queue.
6. If the node is not visited, print it and mark it as visited.
7. Add all adjacent (neighboring) nodes of the current node to the queue.
8. Repeat Steps 5–7 until the queue becomes empty.
9. Stop the program.

Python Code

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

def bfs(start):
    visited = set()
    queue = deque([start])
    while queue:
        node = queue.popleft()
        if node not in visited:
            print(node, end=" ")
            visited.add(node)
```

```
queue.extend(graph[node])

bfs('A')
```

```
EXP 7(BFS Algorithm).py > ...
1  from collections import deque
2  graph = {
3      'A': ['B', 'C'],
4      'B': ['D', 'E'],
5      'C': ['F'],
6      'D': [],
7      'E': [],
8      'F': []
9  }
10 def bfs(start):
11     visited = set()
12     queue = deque([start])
13     while queue:
14         node = queue.popleft()
15         if node not in visited:
16             print(node, end=" ")
17             visited.add(node)
18             queue.extend(graph[node])
19     bfs('A')
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\OPEN CV\AI Lab> & C:\Users\user\AppData\Local\Programs\Python\Python39\python.exe EXP 7(BFS Algorithm).py
A B C D E F
```